

# Juliardi Fajar Hidayatullah

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**Nationality:** Indonesian · **Location:** Indonesia (remote, working with a Dubai-based team)

## ABOUT ME

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I am a Lead AI Platform Engineer and Architect with 10+ years of software and platform engineering experience and, for the last two years, a track record of shipping production agentic AI for enterprise and government clients in the United Arab Emirates, including Dubai Government entities, as part of a Dubai-based team that I work with remotely from Indonesia. I design and build the foundations that let AI agents operate against enterprise systems of record: multi-agent orchestration with MCP tool calling, retrieval and evaluation pipelines, guardrails and audit trails, least-privilege identity, and deployment patterns for on-premises and air-gapped environments where data cannot leave the customer's network.

I have shipped production agent and LLM workloads on Google Vertex AI and Amazon Bedrock, delivered a transactional chatbot on Azure OpenAI / AI Foundry, and served models on-premises with vLLM and Ollama, designing model-agnostic gateways so customers can bring their own model. On the enterprise side I have integrated an ERP end to end: Oracle NetSuite (SuiteTalk REST) for sales orders, invoices and customer payments, feeding a finance reconciliation platform that cut manual reconciliation by 90%.

My foundation was built at Indonesia's leading technology companies (Tokopedia, Ruangguru, Sorabel, Evermos), where I designed company-wide platforms for change data capture, search, messaging, CRM and role-based access control that raised engineering productivity by 70–90%. I build and operate end-to-end MLOps pipelines with Kubeflow, KServe, MLflow, vLLM and Ollama; use Qdrant and pgvector for retrieval-augmented generation; and run evaluation, tracing and monitoring with Langfuse and Arize Phoenix.

I am proficient in Python, Go, Java, JavaScript, PHP and C, with extensive experience in PostgreSQL, MySQL, BigQuery, Redis, Elasticsearch and Kafka, advanced Kubernetes (Helm, Kustomize) and Terraform, and cloud experience across Google Cloud, AWS and Azure. I speak at engineering events on agentic harness design: the tools, memory, control flow and evaluation that turn LLMs into reliable production systems.

## TECHNICAL SKILLS

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- **Agentic AI:** MCP tool calling, A2A agent-to-agent protocol, agent registry and intent routing, RAG (Qdrant, pgvector, Bedrock Knowledge Bases), guardrails and confirmation gates, evaluation and tracing (Langfuse, Arize Phoenix), LangChain, context engineering
- **Cloud AI platforms:** Google Vertex AI (Gemini, tool calling), Amazon Bedrock (Knowledge Bases), Azure OpenAI / AI Foundry, vLLM, Ollama, TGI, Kubeflow, KServe, MLflow
- **Platform and infrastructure:** Kubernetes (Helm, Kustomize, bare-metal and on-prem clusters), Terraform (modular, multi-environment), Docker, CI/CD pipelines, observability and drift / latency dashboards
- **Cloud and deployment models:** Google Cloud, AWS, Azure, on-premises, hybrid and air-gapped deployments, multi-environment rollouts with Helm and Kustomize
- **Data, messaging and ERP integration:** PostgreSQL (pgvector), MySQL, BigQuery, Redis, Elasticsearch, Kafka, Google Pub/Sub, Change Data Capture, Oracle NetSuite (SuiteTalk REST, token-based auth), HubSpot, WhatsApp Business API
- **Languages and APIs:** Python (FastAPI, Flask), Go, Java, JavaScript / React, PHP, C; REST, gRPC, OpenAPI, WebSocket / WebRTC (LiveKit); Clean Architecture, DDD

## WORK EXPERIENCE

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### LEAD AI PLATFORM ENGINEER – ZAFARLABS

01/01/2026 – Current · Dubai, United Arab Emirates (delivered remotely)

As a Forward Deployed Engineer, I embed with enterprise and government clients to design, build and ship agentic generative AI in production.

- **AI Multi-Agent Platform – Dubai Government Services:** built a production multi-agent platform for a Dubai Government entity together with the client's technical architect. We defined the architecture jointly: an A2A (agent-to-agent) envelope protocol for typed message passing, a centralized Agent Registry for discovery and lifecycle, and an Intent Router that dispatches citizen queries to domain agents (traffic and roads, insurance, police fines, community services). I owned the MCP integration for the community services domain and the generic agent all

domain agents are built from; dynamic system prompts and context memorization keep responses grounded and auditable while handling unpredictable citizen queries.

- **AI SOC Platform – Dubai Government security entity:** architected and deployed an air-gapped, on-premises LLM-powered Security Operations Center for real-time threat detection and automated incident triage, integrating asset discovery and third-party security tooling (EDR, SIEM, firewall) into a centralized command centre with human approval on every recommended action.
- **MLOps pipeline – Arabic voice AI:** built an on-premise TTS model versioning pipeline (MLflow) covering experimentation through inference deployment, and delivered an end-to-end Arabic voice customer-service solution: WebSocket / WebRTC real-time audio, Whisper and Google Cloud Speech STT, ElevenLabs and AWS Polly TTS, reducing dependency on human agents.

**Technical environment:** Python (FastAPI), Kubernetes, Helm, Kustomize, Terraform, Docker, MLflow, vLLM, Ollama, LangChain, MCP, Qdrant, PostgreSQL, Redis, Langfuse, Arize Phoenix, LiveKit, Whisper, ElevenLabs

## **AI ARCHITECT & TECHNICAL LEAD – INDEPENDENT (PART-TIME)**

*10/11/2024 – Current · Indonesia*

Part-time architectural guidance and hands-on delivery for early-stage companies moving from AI experimentation to production systems.

- **E-commerce AI assistant on Google Vertex AI:** designed and built a shopping assistant on Gemini through the Vertex AI API with tool calling, answering product questions and making recommendations grounded in the store's catalogue, with conversational flow management and guardrails around what the assistant may say and do.
- **On-premise, air-gapped document analyzer:** built a RAG-based Q&A system over internal company policies that runs entirely inside the firewall with zero internet dependency, proving that secure knowledge retrieval is viable where data cannot leave the building.
- **Business workflow automation with agentic developer tools:** turned processes that previously needed dedicated engineering effort into autonomous, repeatable operations the business runs independently, using a CLI-based agentic harness.

## **LEAD PLATFORM ENGINEER – EVERMOS**

*07/2021 – 05/01/2026 · Bandung, Indonesia*

As lead engineer I owned technical architecture, engineering best practices and the company-wide SaaS platform services, including Kubernetes cluster management, CI/CD pipelines and cloud infrastructure on AWS and GCP.

- **ERP integration – Oracle NetSuite:** integrated the commerce platform with Oracle NetSuite over SuiteTalk REST with token-based authentication for sales orders, invoices and customer payments. Owned the payment contract consumed by the ERP team and partnered with the finance reconciliation team and the finance / ops engineering team. This integration underpinned the Ultimate Finance Tools platform, whose architecture I designed, reducing manual reconciliation by almost 90%.
- **AI social-commerce assistant – Azure OpenAI / AI Foundry:** shipped a chatbot that answers reseller FAQs and creates transactions through tool calling into the order service, with a confirmation step before every write and service identity scoped through the RBAC platform, significantly improving new-user onboarding.
- **Retrieval-augmented generation on Amazon Bedrock:** built RAG on Amazon Bedrock Knowledge Bases to ground assistant answers in company content.
- **RBAC platform:** developed the centralized role-based access control service used by every internal application, with admin UI and security APIs, improving productivity for access-control features by 70%.
- **CDC platform:** created a Kafka-based Change Data Capture service with schema-based setup for real-time data synchronization, improving productivity for CDC features by 85%.
- **Search platform:** built a centralized Elasticsearch-based search service with simple integration APIs, improving productivity for search features by 85%.
- **WhatsApp messaging platform:** built the WhatsApp Business API services used company-wide for OTP, reverse OTP and business communications, improving productivity for messaging features by 90%.
- **CRM platform:** established the customer relationship management platform behind shipping and customer workflows, improving productivity for CRM features by 90%.

**Technical environment:** Go, Python, JavaScript / React, PostgreSQL, Redis, Elasticsearch, Kafka, Docker, Kubernetes, AWS, GCP, Azure OpenAI, Amazon Bedrock, Oracle NetSuite

## SENIOR SOFTWARE ENGINEER – RUANGGURU

05/2020 – 05/2021 · Jogjakarta, Indonesia

- Led the backend team for the first release of the Roboguru chatbot, managing scope and timeline to hit a marketing moment; within a year Roboguru generated the highest number of fresh contacts of any channel.
- Created a shared queue library that replaced Google Pub/Sub for throttling marketing outbound messages, improving robustness and reducing Pub/Sub costs; adopted by other projects.
- Developed the HubSpot integration for the marketing workflow.
- Worked with the VP of Engineering and Engineering Manager to formulate backend design patterns.

## SENIOR SOFTWARE ENGINEER – SORABEL

03/2019 – 05/2020 · Jogjakarta, Indonesia

- Designed BigQuery datasets and schemas and built product features on BigQuery queries.
- Developed and maintained hot-path microservices: coupon, order, product catalogue and the chat platform, including third-party integrations.
- Made the marketing broadcast campaign 9x faster using goroutines with larger payloads; wrote the Medium article "Using Goroutine to Boost Marketing Broadcast Campaign".
- Led the product catalogue v2 project as PIC: initiated codebase and architecture, ran weekly syncs with engineering and product managers, built a reverse proxy for regression control and mentored contributors.

## SOFTWARE ENGINEER – TOKOPEDIA

12/2017 – 01/2019 · Jakarta, Indonesia

- Built the KOL (Key Opinion Leader) recommendation engine on the feeds page using dgraph.
- Built the affiliate tracker engine on PostgreSQL and Redis, and the Feeds and Affiliate APIs over HTTP REST and gRPC.
- Designed the Tokopedia for Business repository and monitored and maintained backend services.

## EDUCATION AND TRAINING

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### BACHELOR OF APPLIED SCIENCE (BASc), COMPUTER ENGINEERING – PENS

10/2012 – 10/2016 · Surabaya, Indonesia · <https://www.pens.ac.id>

## PUBLICATIONS

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**Kinematic Analysis of 7 DoF Head Humanoid FLoW: V-REP Simulation** (2016). Final project at PENS: formulated the mathematical model of a 7-DoF humanoid robot head in the V-REP simulator and controlled it from a desktop application. Presented at the IEEE-sponsored International Electronics Symposium (IES); available on the IEEE portal. Authors: Juliardi Fajar Hidayatullah, Dadet Pramadihanto, Raden Sanggar Dewanto, Ahmad Subhan Khalilullah.

## HONOURS AND AWARDS

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- **05/2017 – PPBT (Pengusaha Pemula Berbasis Teknologi), PENS:** fully funded technology start-up programme with product development, workshops and mentoring.
- **09/2016 – Presenter, International Electronics Symposium (IES), PENS:** presented the FLoW humanoid head kinematics paper in the Robotics and Automation category at an IEEE-sponsored international conference.
- **03/2015 – Finalist, Imagine innovation technology category, Microsoft Indonesia:** technology solution to a city problem built on Microsoft development tools (Visual Studio, C#).